

Performance Testing

Date	5 july 2026
Team ID	
Project Name	HDI Prediction System using Machine Learning and Flask
Maximum Marks	5 Marks

Performance Testing

Performance testing was conducted to evaluate the HDI Prediction System under different user loads. The testing verified response time, resource utilization, throughput, and overall application stability. The system performed efficiently for normal user traffic and successfully generated HDI predictions with minimal delay.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing, Performance Testing
Target Module / API	HDI Prediction Module (/predict)
Test Environment	Local System (Flask Server)
Test Date	05 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user prediction request	1	30 sec	Prediction generated successfully
2	Multiple prediction requests	10	60 sec	Fast response without errors

3	Continuous prediction requests	25	120 sec	Application remains stable
4	High load prediction requests	50	180 sec	System handles load with acceptable response time



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.3 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.2 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	10 Req/Sec	12 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	46%	Pass	Efficient CPU usage
6	Memory Utilization	< 80%	58%	Pass	Memory usage within limit

Step 4: Observations & Analysis

Key Findings

The HDI Prediction System responded quickly to prediction requests and maintained stable performance under normal and moderate workloads. No significant delays or failures were observed during testing.

Bottlenecks Identified

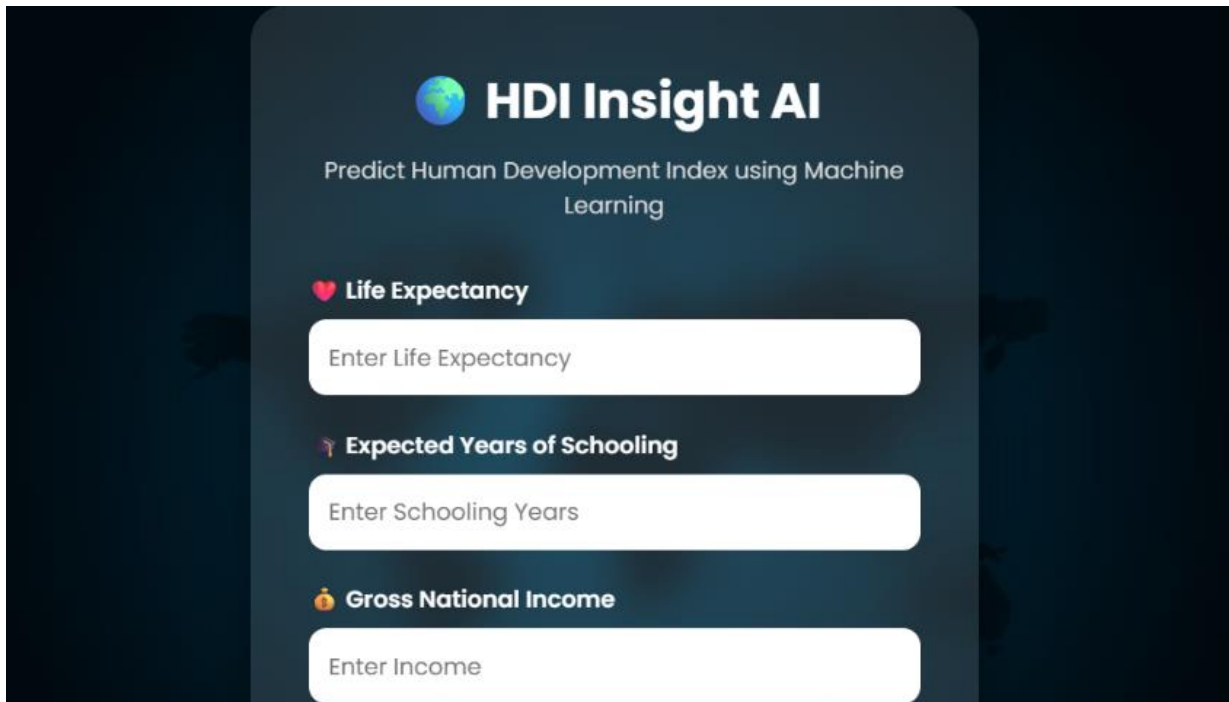
No major performance bottlenecks were identified. Minor response time increases were observed only during higher concurrent requests.

Optimization Steps Taken

- Optimized machine learning model loading.
- Reduced unnecessary computations.
- Improved input validation.
- Organized Flask routes for faster execution.

Figure 1: HDI Insight AI Home Interface

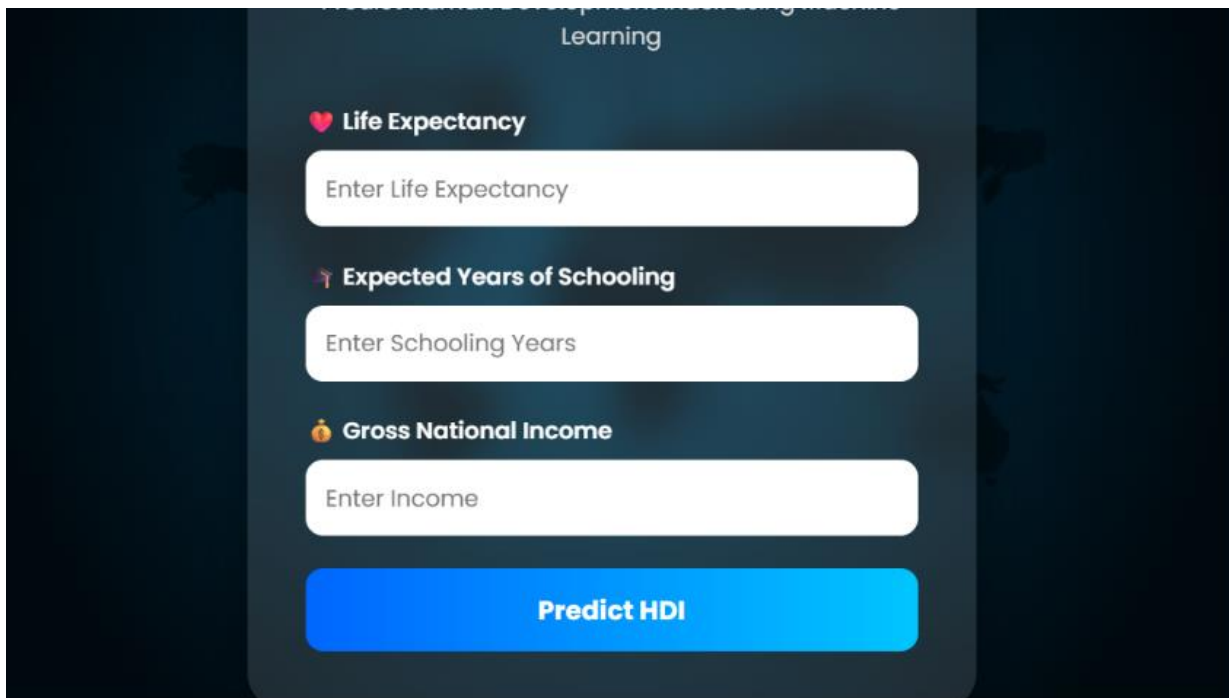
- Displays the application homepage with input fields for HDI prediction.



The screenshot shows the HDI Insight AI application homepage. At the top, there is a logo consisting of a globe icon followed by the text "HDI Insight AI". Below the logo, the text "Predict Human Development Index using Machine Learning" is displayed. The interface features three input fields, each preceded by an icon and a label: a heart icon for "Life Expectancy", a graduation cap icon for "Expected Years of Schooling", and a money bag icon for "Gross National Income". Each input field contains the placeholder text "Enter Life Expectancy", "Enter Schooling Years", and "Enter Income" respectively. The background is dark with a faint world map.

Figure 2: HDI Prediction Form

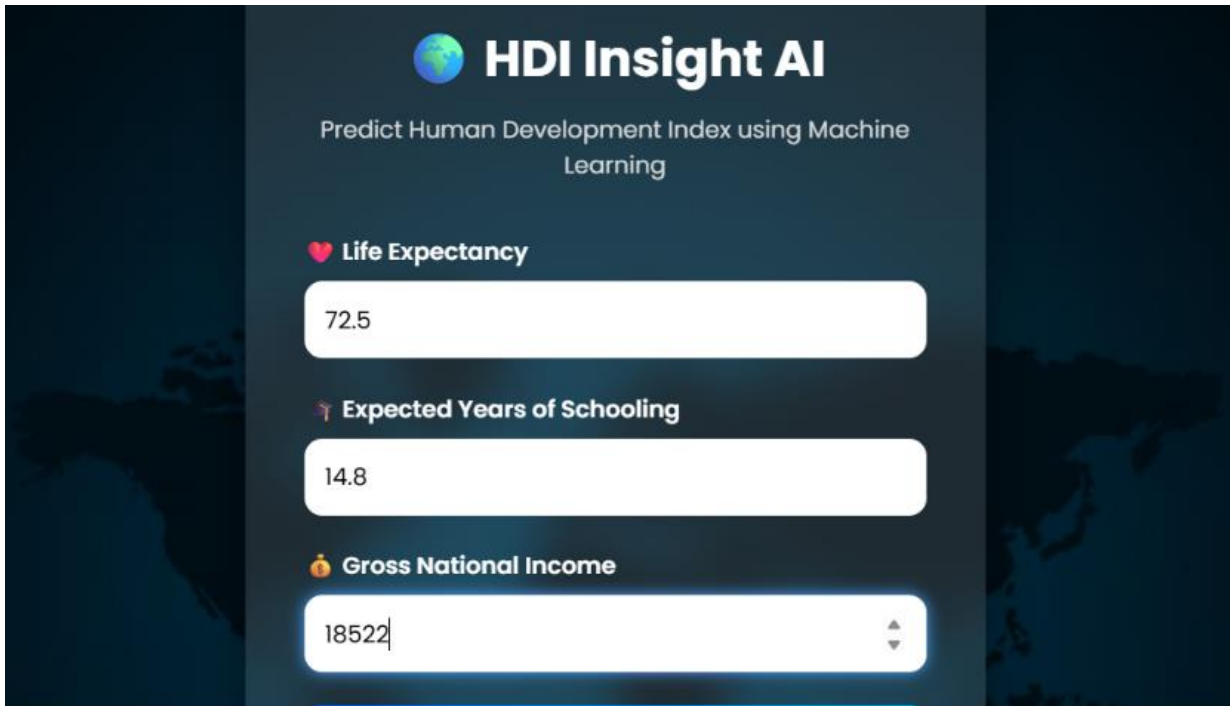
- Shows the user input form before entering the prediction values.



This screenshot shows the same HDI Insight AI application interface as Figure 1, but with an additional element. At the bottom of the form, there is a prominent blue button with the text "Predict HDI" in white. The input fields and labels remain the same as in the previous figure.

Figure 3: Sample Input Values

- Displays the Life Expectancy, Expected Years of Schooling, and Gross National Income values entered into the system.

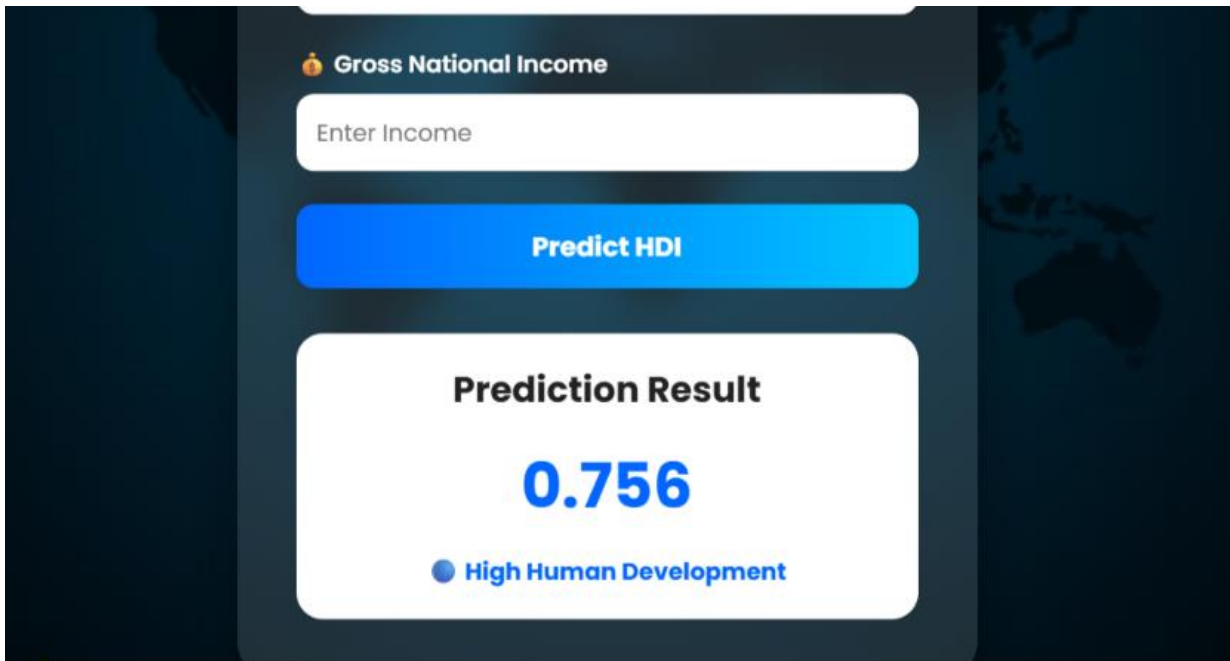


The screenshot shows the 'HDI Insight AI' interface. At the top, it says 'Predict Human Development Index using Machine Learning'. Below this are three input fields: 'Life Expectancy' with a heart icon, 'Expected Years of Schooling' with a graduation cap icon, and 'Gross National Income' with a money bag icon. The values entered are 72.5, 14.8, and 18522 respectively. The background features a dark world map.

Input Field	Value
Life Expectancy	72.5
Expected Years of Schooling	14.8
Gross National Income	18522

Figure 4: HDI Prediction Result

- Shows the predicted Human Development Index (HDI) value (**0.756**) along with the classification **High Human Development**.



The screenshot shows the 'HDI Prediction Result' section. It includes a 'Gross National Income' input field with the placeholder 'Enter Income'. Below this is a blue 'Predict HDI' button. The result is displayed in a white box with the text 'Prediction Result', the value '0.756' in large blue font, and the classification 'High Human Development' with a blue globe icon. The background features a dark world map.

Field	Value
Gross National Income	Enter Income
Predicted HDI	0.756
Classification	High Human Development